$QRTT$: Stateful Round Trip Time Estimation for Wireless Embedded Systems Using Q-Learning

A. B. M. Alim Al Islam and Vijay Raghunathan

***Abstract*—Wireless embedded systems such as sensor nodes and smartphones highlight the importance of reliable data transmission in their advanced applications. Such reliable transmission frequently exhibits a performance below expectation due to inherent resource constraints of the systems. Accurate estimation of round trip time, an important parameter of reliable data transmission, has the potential to significantly improve the performance withstanding the resource constraints. However, contemporary round trip time estimation schemes cannot significantly improve the performance using their stateless estimation schemes as two different states of transmission, *success* and *failure*, are equally prominent over wireless mediums. To address this key fact, we propose a novel stateful round trip time estimation scheme for wireless embedded systems, incorporating a light-weight artificial intelligence method. We apply the proposed scheme in a retransmission timeout mechanism of TCP and compare the efficacy of the scheme against that of contemporary estimation schemes. Exhaustive simulation and a testbed experiment reveal that our scheme significantly improves network throughput and average energy per bit compared to other schemes.**

***Index Terms*—Learning systems, transport protocols, wireless sensor networks.**

I. Introduction

OVER the past few years, wireless embedded systems have attracted, to a great extent, the interest of the research community, resulting in various types of wireless sensor nodes. Recent advanced applications of wireless embedded systems, which frequently demand reliable data transmission, suffer from low performance due to the resource constraints of the systems. The protocols for reliable transmission, such as TCP, utilize estimated round trip time (RTT), which influences the performance while withstanding the constraints. Therefore, suitable estimation of RTT has great potential for enhancing the performance of reliable transmission.

Different techniques [1], [4] have already been proposed for suitable RTT estimation. All of the proposed techniques rely on the experience of previous transmission attempts while performing their estimations. They treat all previous attempts in a similar way, which exhibits good performance in wired networks due to high success rates over wired mediums. However, data transmission over wireless mediums frequently suffers from nondeterministic packet drops, rendering similar treatments inappropriate for wireless embedded systems. Therefore, we segregate the RTT estimation process for these systems into two separate cases: successful deliveries and packet drops. We prefer use of an artificial intelligence (AI)-based scheme in the estimation to mine underlying knowledge of frequently varying RTTs, and thus increase the accuracy of the estimation. We choose a low overhead AI technique called Q-Learning [6] to be consistent with the spirit of applications over resource-constrained wireless embedded systems.

Thus, we make the following contributions in this paper: 1) we present a Q-Learning-based stateful RTT ($QRTT$) estimation technique for wireless embedded systems and exemplify its applicability in TCP; 2) we evaluate $QRTT$ through extensive `ns-2` simulation, which reveals up to 167% and 67% improvement in throughput and average energy per bit, respectively, compared to contemporary estimation techniques; and 3) we perform a testbed experiment for real-time evaluation.

II. Related Work

Different techniques have been proposed to estimate RTT over the Internet, as RTT is a key component of TCP and TCP mostly dominates the Internet. Jacobson *et al.* estimates RTT using a weighted average of samples [1], which is further improved by several research studies [2]. Some other approaches [3] adopt passive estimation of RTT. However, these methods address wired mediums, which significantly differ from the unreliable medium of wireless embedded systems. A few methods [4] attempt to address the unreliability by adjusting Jacobson's approach [1], and thus by treating successful and failed delivery attempts in a similar way. In addition, a few approaches [5] propose a combination of a Kalman filter and cumulative sum to estimate RTT in heterogeneous networks comprising both wired and wireless links. These approaches estimate RTT only upon the arrival of an acknowledgment of a successful attempt, and assume that the impact of failed attempts is accumulated within an increased queuing delay. Thus, these approaches miss the impact of failed attempts that do not increase queuing delays. We can experience such attempts in different situations such as when losses of acknowledgements are not queued.

III. Mechanism of Stateful RTT Estimation

The previous discussion illustrates that we should separately estimate RTTs for successful transmissions and packet drops in wireless networks. Thus, we adopt a low overhead AI technique called Q-Learning for these separate estimations. Next, we provide a brief overview of Q-Learning before elaborating on our estimation technique.

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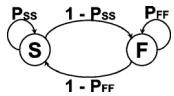

Fig. 1. State transition diagram for a wireless transmission.

A. Q-Learning

Q-Learning is a form of reinforcement learning technique, which optimally takes decisions in domains exhibiting the underlying nature of a Markov Process. Q-Learning experiences the consequence of a decision in the form of a reward and uses a "Reward-Matrix" to contain the rewards received for arriving at a particular state after taking the decision. In addition, it uses a "Q-Matrix," which is updated according to the received reward along with the corresponding state transition. The Q-Matrix guides the determination of a subsequent optimum decision, based on already taken decisions, to maximize the next reward. Also, it occasionally probes unexplored states in a search space to escape local optima.

Q-Learning updates an entry of the Q-Matrix as follows:

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha(s_t, a_t) \times \{r_{t+1} + \gamma \times \max_a Q(s_{t+1}, a) - Q(s_t, a_t)\} \quad (1)$$

where, s_t, a_t, $\alpha(s_t, a_t)$, and r_t are state, action taken, learning rate, and reward at time t, respectively. In addition, γ is a discount factor. Here, the learning rate controls the momentum of the learning process and the discount factor determines the importance of future rewards. In addition to these parameters, the complete Q-Learning process utilizes another parameter ϵ to determine the probability of a random action, which enables avoidance of the phenomenon of being trapped in a local minima.

B. Q-Learning Based Stateful RTT ($QRTT$) Estimation

We estimate RTT for successful and failed transmission attempts in separate manners by adopting two states of data transmission: *success* and *failure*. Reception of an acknowledgment to a sender indicates a successful attempt and moves the sender to a *success* state (S). On the other hand, expiration of the retransmission timer indicates a failed attempt and moves the sender to a *failure* state (F). Fig. 1 shows a corresponding state transition diagram. Here, P_{xy} stands for the probability of a transition from state x to state y. It is worth mentioning that if we would simply segregate Jacobson's approach into two cases, we would miss the impact of cross-transitions, i.e., $x \neq y$, due to its stateless weighted sum technique, which we overcome using Q-Learning in our approach. Here, the usage of Q-Learning only estimates RTT, rather than maximizing any notion of reward similar to the original approach.

The probabilities of all transitions in Fig. 1 solely depend on the corresponding source states, as a success scenario of a transmission over a wireless link only depends on the current status of the link. Therefore, the state transition diagram resembles a Markov process, and thus enables us to utilize Q-Learning to estimate RTTs in two different states. Consequently, we utilize (1) in a slightly modified form as

$$Q(s_t) \leftarrow Q(s_t) + \alpha_{s_t} \times \{d_{t+1} + \gamma_{s_{t+1}} \times Q(s_{t+1}) - Q(s_t)\} \quad (2)$$

Here, t and $t+1$ indicate previous and current states, respectively. Thus, (2) exploits cross-transitions between states. This equation only performs RTT estimation, which is the sole action in our case. Therefore, we use a one-dimensional Q-Matrix in place of a two-dimensional Q-Matrix of (1). Moreover, for the same reason, we take the Q-value at the current state as the multiplier of discount factor. Besides, we adopt measured delay (d_t) as the reward (r_t). Finally, we consider a state-dependent learning rate (γ_{s_t}) in (2) in place of the unique stateless learning rate (γ) in (1) to support diversified environments. For example, in an environment having only one system with a very small number of flows, we should set the value of the discount factor at the *success* state higher than that at the *failure* state. Such a setting increases throughput by increasing the rate of retransmission in the case of failed transmissions. On the other hand, in an environment having multiple interfering systems or having a very large number of cross flows, we should set the value of the discount factor at the *failure* state higher than that at the *success* state to increase throughput by decreasing the extent of congestion. We provide basic guidelines for choosing the values of these parameters. Before presenting the guidelines, we evaluate the efficacy of our proposed estimation technique.

IV. Evaluation of $QRTT$ Using Simulation

We start our evaluation by simulating various network topologies. We choose `ns-2` as our simulation platform due to its wide acceptability in wireless research. In the simulation, we adopt TCP retransmission timeout (RTO) adjustment as our underlying application for being the most prominent among all applications of RTT estimation. The TCP RTO adjustment in `ns-2` implements Jacobson's RTT estimation [1] with the enhancement proposed by Karn [2], which is called smoothed RTT (SRTT). On the other hand, we implement $QRTT$ estimation separately in the TCP code of `ns-2`.

First, we simulate a grid topology with 10×10 wireless sensor nodes having 802.15.4 radio. The nodes are uniformly placed at a distance of 30 m from adjacent nodes in the same row and in the same column. We conduct the simulation with 20 end-to-end cross flows with CBR traffic for 200 s. Each packet in the flow contains 28 bytes of application data. We skip other details on simulation settings due to space limitation.

In our analysis, we collect SRTTs (as RTO/β with $\beta = 2$) and their standard deviations ($StDev$s) for Jacobson's approach, and $QRTT$s with their $StDev$s at both *success* and *failure* states for our approach. Here, SRTT and $QRTT$s in both states are used as the components of RTO calculation. We plot these components of RTO and calculated RTOs for both Jacobson's and our approach in Fig. 2. We plot 200 consecutive values starting from the 100th calculation to make the figure easier to understand. However, all numerical values presented in this paper are calculated over complete simulation runs.

Fig. 2(a) reveals that Jacobson's approach exhibits significantly varying SRTT, with an increasing $StDev$ due to accumulation of information of both *success* and *failure* states into one stateless estimated value. The significantly varying nature would remain even with an improved technique such as Q-Learning due to the stateless estimation approach. On the other hand, Fig. 2(b) reveals that our stateful approach exhibits less varying $QRTT$s in two states with nearly saturated values of $StDev$. Here, $StDev$ of Jacobson's approach (8.75 s) is almost double either of the $StDev$s in our approach (4.88 s in *success* and 4.35 s in *failure*). The smaller values of $StDev$s

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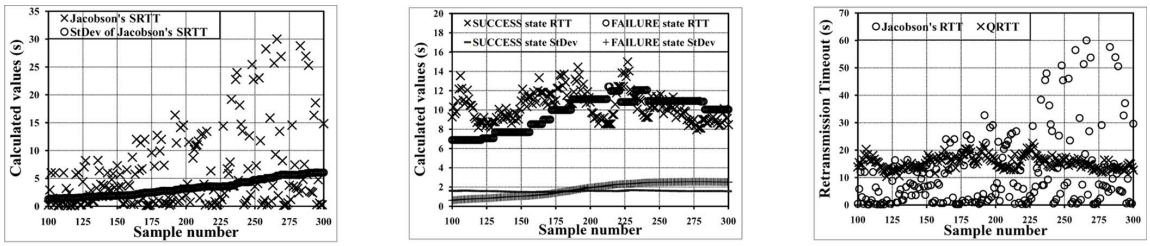
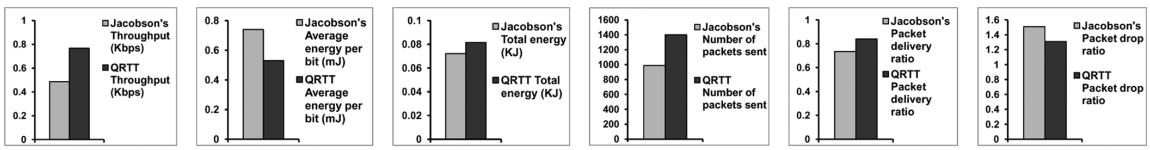


(a) (b) (c)

Fig. 2. Components of RTO and calculated RTOs for Jacobson's RTT and $QRTT$. (a) Jacobson's SRTT with $StDev$. (b) $QRTT$s with $StDev$s in two states. (c) RTOs for Jacobson's RTT and $QRTT$.



(a) (b) (c) (d) (e) (f)

Fig. 3. Performance comparison of Jacobson's RTT and $QRTT$. (a) Throughput. (b) Energy per bit. (c) Total energy. (d) Total sent packets. (e) Delivery ratio. (f) Drop ratio.

TABLE I
PERFORMANCE IMPROVEMENT WITH $QRTT$ IN COMPARISON TO JACOBSON'S RTT AND FALLON'S RTT (WITHIN BRACKETS)

Topology	Coverage area (meter^2)	# of nodes	Node placement	Radio	# of flows	Characteristic of flows	% improvement w.r.t. Jacobson and Fallon	
							Throughput	Average energy per bit
$topo1$	300×300	10×10	uniform	802.15.4	20	fixed cross flow	58 (10)	28 (7)
$topo2$	150×150	5×5	uniform	802.15.4	10	fixed cross flow	75 (154)	33 (45)
$topo3$	60×60	2×2	uniform	802.15.4	4	fixed cross flow	104 (104)	67 (67)
$topo4$	300×300	10×10	random	802.15.4	20	random flow	36 (47)	25 (38)
$topo5$	150×150	5×5	random	802.15.4	10	random flow	26 (41)	16 (60)
$topo6$	60×60	2×2	random	802.15.4	4	random flow	46 (30)	27 (27)
$topo7$	2000×2000	10×10	uniform	802.11	20	fixed cross flow	16 (22)	14 (18)
$topo8$	1000×1000	5×5	uniform	802.11	10	fixed cross flow	16 (9)	12 (9)
$topo9$	400×400	2×2	uniform	802.11	4	fixed cross flow	162 (162)	65 (64)
$topo10$	2000×2000	10×10	random	802.11	20	random flow	31 (38)	7 (22)
$topo11$	1000×1000	5×5	random	802.11	10	random flow	84 (29)	36 (11)
$topo12$	400×400	2×2	random	802.11	4	random flow	164 (167)	49 (61)
$topo13$	4000×2500	40×25	random	802.11g	100	random flow	12 (11)	15 (7)

in $QRTT$ confirm a more stable RTT estimation mechanism in the stateful approach in comparison to that in the stateless Jacobson's approach. The stable stateful approach in turn provides the advantage of achieving more stable RTOs than that obtained by Jacobson's approach. Fig. 2(c) depicts the higher stability of RTOs in our stateful approach.

Now, we compare the performance of $QRTT$ with that of Jacobson's RTT on the basis of network throughput and average energy per bit. Fig. 3(a) and (b) shows improved throughput and average energy per bit in $QRTT$ compared to Jacobson's RTT. We analyze the improvement as follows. $QRTT$ quickly detects failed transmissions in both *success* and *failure* states due to its intelligent RTT estimation mechanism. Therefore, it can send a higher number of packets [see Fig. 3(d)] with higher *delivery ratios* [see Fig. 3(e)] and lower *drop ratios* [see Fig. 3(f)] in comparison to Jacobson's RTT. Here, the delivery ratio is N_r/N_s and the drop ratio is N_d/N_s, where N_s is the number of sent packets, N_r is the number of received packets, and N_d is the number of dropped packets that include both initial and retransmission attempts. The increased delivery ratio leads $QRTT$ to achieve higher throughput [see Fig. 3(a)]. Besides, the decreased drop ratio leads $QRTT$ to achieve lower *unutilized* energy consumption, which eventually results in comparable total energy consumption [see Fig. 3(c)] even with higher numbers of sent and received packets. As a result, $QRTT$ achieves lower energy per bit [see Fig. 3(b)] compared to Jacobson's RTT.

All these impacts result in significant performance improvement using $QRTT$ over a network with fixed cross flows. In addition, we also evaluate $QRTT$ in other network settings with fixed cross flows, random settings with random flows, networks with 802.11 radios, and large-scale networks with 1000 randomly placed nodes. We summarize all our findings in Table I. Here, as small topologies ($topo3$, $topo6$, $topo9$, and $topo12$) having mostly small numbers of hops achieve higher improvement, we believe that $QRTT$ can significantly improve the performances of TCPs that exploit small numbers of hops using intermediate caching (such as DTC [7]).

It is worth mentioning that we take the average of energy per bit over all iterations while presenting the results in random settings. We consider the average as $\left(\sum_{1\le i\le n}(P_i \times \prod_{j\ne i} T_j)\right)/(n \times \prod_{1\le i\le n} T_i)$, where P_i and T_i are the power consumption and throughput at ith iteration respectively, and n is the total number of iterations. Here, throughput in each iteration has its multiplicative effect in the whole denominator but not in the whole numerator. Therefore, if we get significantly decreased throughput in one iteration, this average can significantly increase while retaining a moderate average throughput, which is $(\prod_{1\le i\le n} T_i)/n$. This happens in $topo10$ where we get a moderately improved throughput (31%) with a marginally improved energy per bit (7%) for $QRTT$ in comparison to Jacobson's RTT.

In addition to evaluation against Jacobson's RTT, we also evaluate $QRTT$ against a recent optimization proposed by Fallon *et al.*, [4] over SRTT with a *smoothing factor* of 1.0 and a *delay variance factor* of 0.7. The values within brackets in Table I show improvement over the optimization. Here, we obtain several instances of higher improvement compared to that against Jacobson's RTT. This happens as Fallon's approach

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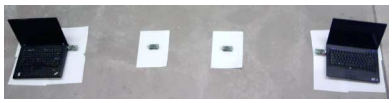
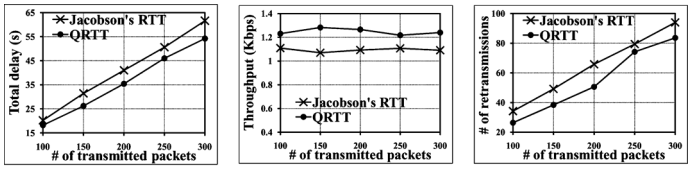


Fig. 4. Testbed setting.



(a) (b) (c)

Fig. 5. Results of testbed evaluation. (a) Total delay. (b) Throughput. (c) Number of retransmissions.

is optimized in a WLAN with no mesh connectivity and no cross-flows, which differ from the characteristics of connectivity and flows in our cases. Finally, we also achieve improved performance with $QRTT$ in large-scale simulation, which we omit due to space limitation.

In our simulation, we select the lower and upper bounds of RTO, learning rate, and discount factor for the Q-Learning. Our lower and upper bounds of RTO in stateful Q-Learning are almost the same as the default values ($\text{lower bound} = 0.2$ s and $\text{upper bound} = 60.0$ s) in Jacobson's estimation in $\text{ns} - 2$, except we make the upper bound 32 s for 802.11 radio for its high bandwidth. Tuning of these bounds is beyond the scope of this paper. On the other hand, we empirically tune the values of the learning rate and discount factor. Tuning of these parameters based on a mathematical model may achieve better performance than that of Table I, which is left as a future work. However, we provide a tentative guideline to adjust the parameters after presenting testbed results in Section V.

V. Testbed Evaluation of $QRTT$

We perform a simple testbed experiment having one flow over 802.15.4 radios to evaluate the impact of $QRTT$. We utilize Telosb motes and program them using TinyOS 2.x to a small transmission power level to run the experiment in our laboratory. The testbed setting is shown in Fig. 4. We enable three-hops stop-and-go flow having 28 bytes of application data in each packet. We connect the source and destination motes to two laptops and control the flow from the laptops.

We transmit different numbers of packets over the motes for five independent runs and take the average of the results. We present the average values in Fig. 5. Fig. 5(a) shows that the total delay is decreased by 14% for $QRTT$ in comparison to Jacobson's RTT, and thus results in 12% improvement in throughput for $QRTT$ [see Fig. 5(b)]. Besides, $QRTT$ achieves an 18% decrease in the number of retransmissions than that with Jacobson's RTT [see Fig. 5(c)], which indicates the potential of lower energy consumption for $QRTT$. These results focus on only one flow, and thus do not incorporate the impact of multiple interfering flows that results in further improvement shown in the simulation results in Section IV.

VI. Guidelines to Choosing Q-Learning Parameters

We use distinct values for four parameters ($\alpha_{success}$, $\alpha_{failure}$, $\gamma_{success}$, $\gamma_{failure}$) in different topologies of our experiment (see Table II). Higher values of these parameters emphasize more recent information, and thus accelerate the learning process. For example, we use larger values for random flows as the average number of hops on a path for a random flow is lower than that for a fixed cross flow. Here, the lower number of hops gives lower delay, and hence more recent information. We also utilize higher values for a high-bandwidth 802.11 radio due to its low latency transmission, which also provides more recent information. Besides, we almost always choose higher values at the *success* state than that at the *failure* state to quickly learn about successful transmissions to enable fast retransmission in case of a failure being at the *success* state. Finally, we use the same values for similar topologies ($topo7 - 9$ and $topo10 - 13$) with the 802.11 radio even though we perform topology-specific tuning for 802.15.4 radio. The reasons behind separate tuning are lower delivery ratio and higher drop ratio for the 802.15.4 radio, which indicates a relatively higher number of failed transmissions than that of the 802.11 radio. Moreover, the ratios for the 802.15.4 radio significantly vary in different topologies, resulting in variation in available information, which also urges for the separate tuning.

TABLE II
Parameters of Q-Learning in $QRTT$ Estimation

Topology	$\alpha_{success}$	$\alpha_{failure}$	$\gamma_{success}$	$\gamma_{failure}$
$topo1$	0.25	0.15	−0.50	0.0075
$topo2$	0.25	0.10	−0.30	0.005
$topo3$	0.25	0.05	−0.50	0.001
$topo4$	0.50	0.50	−0.15	0.125
$topo5$	0.50	0.40	−0.20	0.10
$topo6$	0.70	0.20	−0.30	0.005
$topo7 - 9$	0.50	0.15	−0.50	0.05
$topo10 - 13$	0.90	0.10	−0.90	0.10

VII. Conclusion

Efficient RTT estimation exhibits a crucial impact in performance improvement of TCP over wireless embedded systems. However, current RTT estimation approaches fail to model the unreliable nature of wireless mediums in which the systems operate. Therefore, we propose a novel stateful Q-Learning based RTT ($QRTT$) estimation approach to model the unreliability. We demonstrate better performance of $QRTT$ through testbed experiment and extensive simulation, which uncover the efficacy of $QRTT$ to significantly improve network throughput (up to 167%) and energy per bit (up to 67%) compared to contemporary RTT estimation approaches.

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